Expanding Brackets

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 67 minutes · Total marks: 67

KEY RULES

Single bracket	Multiply every term inside: $3x(2x-5) = 6x^2 - 15x$
Double bracket	$(x+a)(x+b) = x^2 + (a+b)x + ab$
Three brackets	Expand two first, then multiply by the third
Signs	Be careful with negatives: $-(x-3) = -x+3$

WORKED EXAMPLE 1

[2 marks]

Expand $3x(2x-5)$

Step 1	Multiply each term inside by $3x$	$3x \times 2x = 6x^2$
Step 2	Second term	$3x \times (-5) = -15x$

Answer: $6x^2 - 15x$

WORKED EXAMPLE 2

[2 marks]

Expand and simplify $(x+4)(x-3)$

Step 1	Multiply every term by every term	$x^2 - 3x + 4x - 12$
Step 2	Collect like terms	$x^2 + x - 12$

Answer: $x^2 + x - 12$

WORKED EXAMPLE 3

[3 marks]

Expand and simplify $(2x+1)(x-5)$

Step 1	Multiply every term by every term	$2x^2 - 10x + x - 5$
Step 2	Collect like terms	$2x^2 - 9x - 5$

Answer: $2x^2 - 9x - 5$

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1 Expand $4(x+5)$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

2 Expand $3(2y-7)$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

3 Expand $x(x+8)$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

4 Expand and simplify $3(x+2)+2(x-5)$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

5 Expand and simplify $5(2x-1)-3(x+4)$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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6 Expand and simplify $(x+3)(x+7)$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

7 Expand and simplify $(x-4)(x+9)$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

8 Expand and simplify $(x-5)(x-2)$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

9 Expand and simplify $(x+6)^2$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

10 Expand and simplify $(2x-3)^2$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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11 Expand and simplify $(x+5)(x-5)$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

12 Expand and simplify $(3x+2)(2x-5)$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

13 Simplify $(2x+3)^2-(2x+3)(x-5)$

Give your answer in the form ax^2+bx+c

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2017 · Question 9

14 Expand and simplify $3x(2x+5)(7x-4)$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 12

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15 Expand and simplify $3x(2x - 5)^2$
Show clear algebraic working.

.....

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 13

16 Expand and simplify $4x(3x + 1)(2x - 3)$
Show your working clearly.

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 13

17 Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 14

18 Write $3x(2x-1)(5x+4)$ in the form $ax^3 + bx^2 + cx$ where a, b and c are integers

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 15

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CHALLENGE QUESTIONS

19 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 16

20 (a) Expand and simplify $5x(x + 2)(3x - 4)$

.....

(3)

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

.....

(3)

(Total for question = 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2022 · Question 13

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Expanding Brackets

21 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

.....

(3)

(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

$y =$ .....

(3)

(Total for question = 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 14

22 (a) Expand and simplify $(3x + 1)(2 - x)(4 + x)$

.....

(3)

b) Simplify fully $\left(\frac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$

.....

(3)

(Total for Question 14 is 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 14

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23 (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$

.....

(3)

b) Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$

.....

(3)

(Total for Question 16 is 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 16

TOTAL FOR PAPER IS 67 MARKS

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